

Checkpoint F5: Green's theorem

Find $\oint_C \vec{F} \cdot d\vec{r}$, where

$$\vec{F}(x, y) = \langle e^{x^2} - xy, e^{y^2} - xy \rangle$$

and C is the rectangle with corners $(2, 1)$ and $(5, 4)$. (Draw a picture!)